Lec 36

Convergence of RVs. $X_1, X_2, \dots$ be sequence of RVs.

① Almost-sure convergence (SLLN). $X_n \xrightarrow{a.s} X$ $\quad P(\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1$

$$\Downarrow \qquad \sum_{n=1}^{\infty} P(A_n) < \infty \Longrightarrow P(\limsup A_n) = 0 \qquad \Uparrow$$

② Convergence in probability (WLLN) $\quad X_n \to X$ in probability $\quad \lim_{n\to\infty} P(\underbrace{|X_n - X| > \varepsilon}_{A_n}) = 0$

$\Downarrow$ (today)

③ Convergence in distribution (Central limit theorem). $\quad X_n \to X$ in distribution $\quad \lim_{n\to\infty} F_{X_n}(x) = F_X(x)$

$X_n = U$

$X = 1-U$.

$X_n \not\to X$ probability

$X_n \to X$, distribution.

$X_1, X_2, \dots$ iid RVs. $E[X_i] = \mu$, $Var(X_i) = \sigma^2$

$$Z_n = \frac{\sum_{i=1}^{n}(X_i - \mu)}{\sqrt{n}\,\sigma} \qquad E[Z_n] = 0,\ Var(Z_n) = 1.$$

$Z_n \longrightarrow Z$ in distribution sense $\quad Z \sim N(0,1)$

$\lim_{n\to\infty} F_{Z_n}(z) = F_Z(z)$

$\Updownarrow$

$\lim_{n\to\infty} \log M_{Z_n}(s) = \log M_Z(s)$

$\bar{X}_i = \frac{X_i - \mu}{\sigma}$, $E[\bar{X}_1^2] = \frac{E[(X_1-\mu)^2]}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1$

$\phi_X(t) = E[e^{itx}]$

$M_Z(s) = E[e^{Zs}]$

$\log M_{Z_n}(s) = \log E[e^{\sum_{i=1}^n \frac{\bar{X}_i s}{\sqrt{n}}}]$

$= n \log E[e^{\frac{\bar{X}_1 s}{\sqrt{n}}}]$

$= n \log M_{\bar{X}_1}\left(\frac{s}{\sqrt{n}}\right)$

$\lim_{n\to\infty} \log M_{Z_n}(s) = \lim_{n\to\infty} \frac{\log M_{\bar{X}_1}\left(\frac{s}{\sqrt{n}}\right)}{1/n}$

$= \lim_{n\to\infty} \frac{L\left(\frac{s}{\sqrt{n}}\right)}{1/n}$

$L(s) = \log M_{\bar{X}_1}(s)$

$L'(s) = \frac{1}{M_{\bar{X}_1}(s)} M'_{\bar{X}}(s)$

$L'(0) = \frac{M'_{\bar{X}_1}(0)}{M_{\bar{X}_1}(0)}$

$= \frac{E[\bar{X}_1]}{1} = 0$

$L''($

L''

$$= \lim_{n\to\infty} \frac{L'\left(\frac{s}{\sqrt{n}}\right) s \left(\frac{-1}{2}\right) \cancel{n^{-3/2}}}{-\cancel{n^{-2}}\ n^{-1/2}}$$

$$= \lim_{n\to\infty} \frac{s\, L'\left(\frac{s}{\sqrt{n}}\right)}{2\ n^{-1/2}}$$

$$= \lim_{n\to\infty} \frac{s\, L''\left(\frac{s}{\sqrt{n}}\right) s\ \cancel{\frac{-1}{2}} \cancel{n^{-3/2}}}{2\times \cancel{\left(\frac{-1}{2}\right)} \cancel{n^{-3/2}}}$$

$$= \lim_{n\to\infty} \frac{s^2}{2} L''\left(\frac{s}{\sqrt{n}}\right)$$

$$L''(s) = \frac{-1}{M_{X_1}(s)^2} \left(M'_X(s)\right)^2 + \frac{1}{M_{X_1}(s)} M''_X(s)$$

$$L''(0) = -E[X_1]^2 + \frac{E[X_1^2]}{1}$$

$$\Rightarrow L''(0) = 1.$$

$$\lim_{n\to\infty} \log M_{Z_n}(s) = \frac{s^2}{2}$$

$$= \log M_Z(s)$$

$$Z \sim N(0,1) \Longrightarrow M_Z(s) = e^{s^2/2}$$

Berry Essen Theorem

Let $X_1, X_2, \ldots$ are iid

$$Z_n = \frac{\sum_{i=1}^{n} (X_i - \mu)}{\sigma\sqrt{n}}$$

$$\left| F_{Z_n}(z) - F_Z(z) \right| \le \frac{C\rho}{\sigma^3\sqrt{n}}$$

where

$\Rightarrow L''(0) = 1.$

$$\lim_{n \to \infty} \log M_{Z_n}(s) = \frac{s^2}{2}$$

$$= \log M_Z(s)$$

$$Z \sim N(0,1) \Longrightarrow M_Z(s) = e^{s^2/2}$$

Berry Essen Theorem

Let $X_1, X_2, \ldots$ are iid.

$$Z_n = \frac{\sum_{i=1}^{n} (X_i - \mu)}{\sigma \sqrt{n}}$$

$$\left| F_{Z_n}(z) - F_Z(n) \right| \le \frac{C \rho}{\sigma^3 \sqrt{n}}$$

where $\rho = E[|X|^3] < \infty$

and $\sigma^2 = E[X^2] < \infty$

and $E[X_1] = 0$.

$$C \ge \frac{1}{\sqrt{2\pi}} + 0.011$$

Lec 36

Example: Exit Poll"

Let p be the fraction of voters that support Trump for office. n "randomly "selected" voters are interviewed & M_n fraction of them support him.

Each voter's response is $X_i \sim Ber(p)$.

$$M_n = \frac{\sum_{i=1}^{n} X_i}{n}$$ How accurately does M_n predict "p"

We want to say with "high confidence" that $M_n \in [p-\epsilon, p+\epsilon]$.

$$P(|M_n - p| \leq \epsilon) \geq 1-\delta$$

Suppose we want to declare that exit poll is 75% confident that the estimate is upto 10^{-2} error. What should sample size of the poll be?

$P(|M_n - p| > \varepsilon) \leq \delta$

$\delta = 1 - 0.95 = 0.05$

$\epsilon = 10^{-2}$

Chebyshev inequality

$P(|M_n - p| > \varepsilon) \leq \frac{Var(M_n)}{\varepsilon^2}$

$= \frac{Var(X_1)}{n\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}$

$Var(X_1) = E[X^2] - (E[X])^2$

$\leq E[cX] - (E[X])^2$

$= E[X](c - E[X])$

If $\frac{1}{4n\varepsilon^2} \leq 0.05$ then we are gaurantied

that $P(|M_n - p| > \varepsilon) \leq 0.05$

(Exercise
Check what gaurantees
Chernoff gives)

$n \geq \frac{1}{4 \times 0.05 \times \varepsilon^2} = \frac{1}{4 \frac{1}{20} \times 10^{-4}} = 20 \times 10^4 \times \frac{1}{4} = 5 \times 10^4$

Using CLT's approximation

$$P(|M_n - p| > \varepsilon) \leq \delta$$

$$= P\left(\left| \frac{\sum_{i=1}^{n} (X_i - p)}{n} \right| > \varepsilon \right)$$

$$W_n = \frac{\sum_{i=1}^{n} (X_i - p)}{n}$$

$$E[W_n] = 0, \quad Var(W_n) = \frac{\cancel{n}\, \sigma^2}{n^{\cancel{2}}}$$

$$= P(|W_n| > \varepsilon)$$

$$= P(W_n > \varepsilon) + P(W_n < -\varepsilon)$$

$$= P\left(\frac{W_n}{\sigma/\sqrt{n}} > \frac{\varepsilon}{\sigma/\sqrt{n}} \right) + P\left(\frac{W_n}{\sigma/\sqrt{n}} < -\frac{\varepsilon\sqrt{n}}{\sigma} \right)$$

$$= 1 - \Phi\left(\frac{\sqrt{n}\,\varepsilon}{\sigma}\right) + \Phi\left(\frac{-\varepsilon\sqrt{n}}{\sigma}\right)$$

$$= 2\left(1 - \Phi\left(\frac{\sqrt{n}\,\varepsilon}{\sigma}\right)\right) \quad \text{as } \Phi(-x) = 1 - \Phi(x)$$

$p = \frac{1}{2}$

Suppose we choose n st

$$2\left(1 - \Phi\left(\frac{\sqrt{n} \times 10^{-2}}{\sigma}\right)\right) \leq 0.05$$

$$1 - 0.025 = \frac{2 - 0.05}{2} \leq \Phi\left(\frac{\sqrt{n} \times 10^{-2}}{\sigma}\right)$$

$$\boxed{n \geq 9604}$$

Lec 36

Convergence in probability $\Longrightarrow$ convergence in distribution.

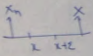
$X_n \longrightarrow X$. in probability.

Given $\lim\limits_{n\to\infty} P\left(|X_n - X| > \varepsilon\right) = 0$.

To show $\lim\limits_{n\to\infty} F_{X_n}(x) = F_X(x)$

$A = \{X \le x+\varepsilon\}$

$F_{X_n}(x) = P(X_n \le x)$



$= P(\{X_n \le x\} \cap A) + P(\{X_n \le x\} \cap A^c)$

$\Rightarrow X - X_n > \varepsilon$ $\quad = P(X_n \le x, X \le x+\varepsilon) + P(X_n \le x, X > x+\varepsilon)$

$\le P(X \le x+\varepsilon) + P(X - X_n > \varepsilon)$

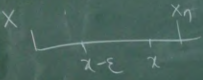
$$F_{X_n}(x) \le F_X(x+\varepsilon) + P(|X - X_n| \ge \varepsilon)$$

$$\lim_{n\to\infty} F_{X_n}(x) \le F_X(x+\epsilon) + \lim_{n\to\infty} P(|X - X_n| \ge \varepsilon)$$

$$= F_X(x+\varepsilon) \quad \text{for any } \epsilon > 0$$

To show that $F_X(x-\varepsilon) \le \lim_{n\to\infty} F_{X_n}(x)$ $\qquad B = \{X \le x-\varepsilon\}$

$$1 - F_{X_n}(x) = P(X_n > x) = P(X_n > x, X \le x-\varepsilon) + P(X_n > x, X > x-\varepsilon)$$



$$\le P(X_n - X > \varepsilon) + P(X > x-\varepsilon)$$

$$\le P(|X_n - X| > \varepsilon) + 1 - F_X(x-\varepsilon)$$

$$\Rightarrow F_{X_n}(x) \ge F_X(x-\varepsilon) - P(X_n - X > \varepsilon)$$

$$\lim_{n\to\infty} F_{X_n}(x) \geq F_X(x-\varepsilon) - \underbrace{\lim_{n\to\infty} P(|X_n - X| > \varepsilon)}_{=0}$$

$$F_X(x-\varepsilon) \leq \lim_{n\to\infty} F_{X_n}(x) \leq F_X(x+\varepsilon).$$

$\lim_{\varepsilon\to 0}$ on all the three quantities gives

$$F_X(x^-) \leq \lim_{n\to\infty} F_{X_n}(x) \leq F_X(x).$$

Assuming CDF of X is continuous at x.

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

$\leq x - \varepsilon\}$

$\varepsilon)$.

$x-\varepsilon)$

$\varepsilon)$